

Factorization

Factorization of $x^n - 1$

Consider the equation $x^n - 1 = 0$.

$$\Rightarrow x^n = 1.$$

$$x^n = \cos 0 \pm i \sin 0$$

In general, $x^n = \cos 2r\pi \pm i \sin 2r\pi$, r is any integer

$$x = (\cos 2r\pi \pm i \sin 2r\pi)^{1/n}$$

$$x = \cos \frac{2r\pi}{n} \pm i \sin \frac{2r\pi}{n} \quad \text{--- (1)}$$

Case 1:- When n is even.

$$\Rightarrow \frac{n}{2} \text{ is an integer.}$$

In (1), Put $r = 0, 1, 2, \dots, \frac{n}{2} - 1, \frac{n}{2}$.

$$\text{When } r = 0, x = \cos 0 \pm i \sin 0 = 1$$

$$r = 1, x = \cos \frac{2\pi}{n} \pm i \sin \frac{2\pi}{n}$$

$$r = 2, x = \cos \frac{4\pi}{n} \pm i \sin \frac{4\pi}{n}$$

$$\dots \dots \dots r = \frac{n}{2} - 1, x = \cos \frac{(n-2)\pi}{n} \pm i \sin \frac{(n-2)\pi}{n}$$

$$r = \frac{n}{2}, x = \cos \pi \pm i \sin \pi = -1$$

These are the n solutions of the eqn $x^n - 1 = 0$.

Thus corresponding to $x=0$ and $x=n/2$, we've solutions, $x=1$ and $x=-1$. These solutions give us linear factors, $x-1$ and $x+1$. Combining them we obtain the quadratic factor x^2-1 .

Corresponding to $x=1$, we've the solutions,

$$x = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n} \quad \text{and} \quad x = \cos \frac{2\pi}{n} - i \sin \frac{2\pi}{n}$$

These solutions give us the linear factors,

$$x - \cos \frac{2\pi}{n} - i \sin \frac{2\pi}{n} \quad \text{and} \quad x - \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$$

Combining them we've the quadratic factors,

$$\left(x - \cos \frac{2\pi}{n} - i \sin \frac{2\pi}{n}\right) \left(x - \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}\right)$$

$$= x^2 - 2x \cos \frac{2\pi}{n} + \cos^2 \frac{2\pi}{n} + \sin^2 \frac{2\pi}{n}$$

$$= x^2 - 2x \cos \frac{2\pi}{n} + 1$$

Similarly corresponding to $x=2, 3, \dots, n/2-1$, we obtain the quadratic factors,

$$\left(x^2 - 2x \cos \frac{4\pi}{n} + 1\right), \left(x^2 - 2x \cos \frac{6\pi}{n} + 1\right), \dots, \left(x^2 - 2x \cos \frac{(n-2)\pi}{n} + 1\right)$$

$$\therefore x^n - 1 = K(x^2 - 1) \left(x^2 - 2x \cos \frac{2\pi}{n} + 1\right) \dots \left(x^2 - 2x \cos \frac{(n-2)\pi}{n} + 1\right)$$

Equating coefficients of x^n ,

we get $K=1$

$\therefore$ When n is even,

$$x^n - 1 = (x^2 - 1) \left(x^2 - 2x \cos \frac{2\pi}{n} + 1 \right) \dots \left(x^2 - 2x \cos \frac{(n-2)\pi}{n} + 1 \right)$$

$$= (x^2 - 1) \prod_{r=1}^{\frac{n}{2}-1} \left(x^2 - 2x \cos \frac{2r\pi}{n} + 1 \right)$$

Case II: When n is odd.

$n-1$ is even and therefore $\frac{1}{2}(n-1)$ is an integer.

In ①, put $r=0, 1, 2, \dots, \frac{n-1}{2}$

When $r=0$, $x = \cos 0 + i \sin 0 = 1$

$r=1$, $x = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$

$r=2$, $x = \cos \frac{4\pi}{n} + i \sin \frac{4\pi}{n}$

$\therefore r = \frac{n-1}{2}$, $x = \cos \frac{(n-1)\pi}{n} + i \sin \frac{(n-1)\pi}{n}$

corresponding to $r=0$, we've the solus $x=1$, this leads to the linear factor $x-1$.

Also as in case I, corresponding to $r=1, 2, 3, \dots, \frac{1}{2}(n-1)$ we've the quadratic factors,

$$\left(x^2 - 2x \cos \frac{2\pi}{n} + 1 \right), \dots, \left(x^2 - 2x \cos \frac{(n-1)\pi}{n} + 1 \right)$$

$$\therefore x^n - 1 = k(x-1) \left(x^2 - 2x \cos \frac{2\pi}{n} + 1 \right) \dots \left(x^2 - 2x \cos \frac{(n-1)\pi}{n} + 1 \right)$$

Equating coeffs of x^n ,

$$1 + \frac{x(1-x)}{n} + \frac{x^2(1-x)^2}{n^2} + \dots + \frac{x^{n-1}(1-x)^{n-1}}{n^{n-1}} = 1 - x^n$$

$\therefore$ When n is odd,

$$x^{n-1} = (x-1) \left(x^2 - 2x \cos \frac{2\pi}{n} + 1 \right) \dots \left(x^2 - 2x \cos \frac{(n-1)\pi}{n} + 1 \right)$$

$$= (x-1) \prod_{r=1}^{n-1} \left(x^2 - 2x \cos \frac{2r\pi}{n} + 1 \right)$$

$$x^n - 1 = (x-1) \prod_{r=1}^{n-1} \left(x^2 - 2x \cos \frac{2r\pi}{n} + 1 \right)$$

Q. Factorise into real factors.

i) $x^n - 1$

$$x^n - 1 = 0$$

$$\Rightarrow x^n = 1$$

$$x^n = \left(\cos(2r\pi) \pm i \sin(2r\pi) \right)^{1/n}$$

$$x = \left(\cos 2r\pi \pm i \sin 2r\pi \right)^{1/n}$$

$$= \cos \frac{2r\pi}{n} \pm i \sin \frac{2r\pi}{n} \quad \text{--- (1)}$$

$$n=7 \text{ is odd} \Rightarrow \frac{n-1}{2} = 3$$

In (1), Put $r=0, 1, 2, 3$

When $r=0$, $x = \cos 0 \pm i \sin 0 = 1$

$r=1$, $x = \cos \frac{2\pi}{7} \pm i \sin \frac{2\pi}{7}$

$r=2$, $x = \cos \frac{4\pi}{7} \pm i \sin \frac{4\pi}{7}$

$$r=3, \quad \omega = \cos \frac{6\pi}{7} \pm i \sin \frac{6\pi}{7}$$

corresponding $r=0$, we've the linear factor $x-1$.

" $r=1$, we've the factors

$$\left(x - \cos \frac{2\pi}{7} - i \sin \frac{2\pi}{7}\right) \text{ and } \left(x - \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}\right)$$

combining these,

$$\left(x - \cos \frac{2\pi}{7} - i \sin \frac{2\pi}{7}\right) \left(x - \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}\right)$$

$$= x^2 - 2x \cos \frac{2\pi}{7} + 1$$

Similarly corresponding to $r=2$ and 3 , we've,

$$\left(x^2 - 2x \cos \frac{4\pi}{7} + 1\right), \left(x^2 - 2x \cos \frac{6\pi}{7} + 1\right)$$

$$\therefore x^7 - 1 = (x-1) \left(x^2 - 2x \cos \frac{2\pi}{7} + 1\right) \left(x^2 - 2x \cos \frac{4\pi}{7} + 1\right)$$

$$\left(x^2 - 2x \cos \frac{6\pi}{7} + 1\right)$$

$$= (x-1) \prod_{r=1}^3 \left(x^2 - 2x \cos \frac{2r\pi}{7} + 1\right)$$

$$\text{ii) } x^{10} - 1 = \dots \left(x^{10} - 1\right) = 0$$

Answer

$$x^{10} = 1$$

$$x = \left(\cos \frac{2\pi}{10} \pm i \sin \frac{2\pi}{10}\right) \text{ --- (1)}$$

$$n=10 \text{ even, } \frac{n}{2} = 5$$

Put $\alpha = 0, 1, 2, 3, 4, 5$

$$\alpha = 0, \quad \alpha = \cos 0 \pm i \sin 0 = 1$$

$$\alpha = 1, \quad \alpha = \cos \frac{2\pi}{10} \pm i \sin \frac{2\pi}{10}$$

$$\vdots$$

$$\alpha = 5, \quad \alpha = \cos \pi + i \sin \pi = -1$$

For $\alpha = 0, \alpha = 5$, we've linear factors $(x-1)$ and $(x+1)$.

$$\Rightarrow (x-1)(x+1)$$

$$\alpha = 1 \Rightarrow \text{we've } x - \cos \frac{2\pi}{10} - i \sin \frac{2\pi}{10} \text{ and } x - \cos \frac{2\pi}{10} + i \sin \frac{2\pi}{10}$$

$\Rightarrow$ we've the factor

$$(x^2 - 2x \cos \frac{2\pi}{10} + 1)$$

Now for $\alpha = 2, 3, 4$ we've

$$(x^2 - 2x \cos \frac{4\pi}{10} + 1), (x^2 - 2x \cos \frac{6\pi}{10} + 1), (x^2 - 2x \cos \frac{8\pi}{10} + 1)$$

$$\therefore x^{10} - 1 = (x^2 - 1) (x^2 - 2x \cos \frac{2\pi}{10} + 1) \dots (x^2 - 2x \cos \frac{8\pi}{10} + 1)$$

$$= (x^2 - 1) \prod_{\alpha=1}^4 (x^2 - 2x \cos \frac{2\alpha\pi}{10} + 1)$$

$$= \left(\frac{x^5 - 1}{x - 1} \right) \dots$$

Factorisation of x^{n+1}

$$x^{n+1} = 0$$

$$x^n = -1$$

$$= \cos \pi \pm i \sin \pi$$

$$= \cos (2r+1)\pi \pm i \sin (2r+1)\pi, \quad r \in \mathbb{Z}.$$

$$\Rightarrow x = \cos \frac{(2r+1)\pi}{n} \pm i \sin \frac{(2r+1)\pi}{n} \quad \text{--- (1)}$$

Case 1: n is even.

$$\Rightarrow n/2 \in \mathbb{Z}.$$

In (1), Put $r = 0, 1, 2, \dots, n/2 - 1$.

$$r=0, \quad x = \cos \frac{\pi}{n} \pm i \sin \frac{\pi}{n}$$

$$r=1, \quad x = \cos \frac{3\pi}{n} \pm i \sin \frac{3\pi}{n}$$

$\vdots$

$$r = \frac{n}{2} - 1, \quad x = \cos \frac{(n-1)\pi}{n} \pm i \sin \frac{(n-1)\pi}{n}$$

These are the n solutions of the eqn $x^{n+1} = 0$ corresponding to $r=0$, we've the solutions,

$$x = \cos \frac{\pi}{n} + i \sin \frac{\pi}{n}, \quad x = \cos \frac{\pi}{n} - i \sin \frac{\pi}{n}.$$

These solutions gives us the linear factors.

$$\left(x - \cos \frac{\pi}{n} - i \sin \frac{\pi}{n}\right)^n \text{ and } \left(x - \cos \frac{\pi}{n} + i \sin \frac{\pi}{n}\right)^n$$

combining, we get.

$$x^2 - 2x \cos \frac{\pi}{n} + 1.$$

By comparing to $x^2 - 2x \cos \frac{\pi}{n} + 1$, we've

$$x^2 - 2x \cos \frac{3\pi}{n} + 1, x^2 - 2x \cos \frac{5\pi}{n} + 1, \dots, x^2 - 2x \cos \frac{(n-1)\pi}{n} + 1.$$

$$\therefore x^n + 1 = K \cdot (x^2 - 2x \cos \frac{\pi}{n} + 1) \cdot \dots \cdot (x^2 - 2x \cos \frac{(n-1)\pi}{n} + 1)$$

Equating the coeff of x^n , $K=1$

$$\begin{aligned} \therefore x^n + 1 &= (x^2 - 2x \cos \frac{\pi}{n} + 1) \cdot \dots \cdot (x^2 - 2x \cos \frac{(n-1)\pi}{n} + 1) \\ &= \prod_{r=0}^{\frac{n}{2}-1} (x^2 - 2x \cos \frac{(2r+1)\pi}{n} + 1) \end{aligned}$$

Case II: when n is odd:

$\Rightarrow n-1$ is even

$$\Rightarrow \frac{n-1}{2} \in \mathbb{Z}.$$

In ①, Put $r = 0, 1, 2, \dots, \frac{n-1}{2}$.

$$\text{When } r=0, x = \cos \frac{\pi}{n} \pm i \sin \frac{\pi}{n}.$$

$$r=1, x = \cos \frac{3\pi}{n} \pm i \sin \frac{3\pi}{n}$$

$$r = \frac{n-1}{2}, x = \cos \pi \pm i \sin \pi = -1$$

corresponding to $r = \frac{n-1}{2}$, we've the linear factor

Also as in case 1, for $r = 0, 1, 2, \dots, \frac{n-3}{2}$, we've quad. factors,

$$(x^2 - 2x \cos \frac{\pi}{n} + 1), (x^2 - 2x \cos \frac{2\pi}{n} + 1), \dots, (x^2 - 2x \cos \frac{(n-2)\pi}{n} + 1)$$

$$\therefore (x^n + 1) = K(x+1)(x^2 - 2x \cos \frac{\pi}{n} + 1) \dots (x^2 - 2x \cos \frac{(n-2)\pi}{n} + 1)$$

equating coeff of x^n , $K=1$.

$$\therefore x^n + 1 = (x+1) \prod_{r=0}^{\frac{n-3}{2}} (x^2 - 2x \cos \frac{(2r+1)\pi}{n} + 1)$$

6. Factorise.

i) $x^6 + 1$

$$x = \cos \frac{(2r+1)\pi}{6} \pm i \sin \frac{(2r+1)\pi}{6} \quad \text{--- (1)}$$

$$n=6, \quad \frac{n-1}{2} = 2.$$

Put $r = 0, 1, 2$ in (1).

When $r=0$, $x = \cos \frac{\pi}{6} \pm i \sin \frac{\pi}{6}$

$r=1$, $x = \cos \frac{3\pi}{6} \pm i \sin \frac{3\pi}{6}$

$r=2$, $x = \cos \frac{5\pi}{6} \pm i \sin \frac{5\pi}{6}$

corresponding $r=0$, we've the factors:

$$(x - \cos \pi/6 + i \sin \pi/6) \text{ and } (x - \cos \pi/6 - i \sin \pi/6).$$

$$\Rightarrow (x - \cos \pi/6 + i \sin \pi/6) (x - \cos \pi/6 - i \sin \pi/6)$$

$$= x^2 - 2x \cos \pi/6 + 1.$$

Similarly for $r=1, 2$,

$$x^2 - 2x \cos 3\pi/6 + 1, \quad x^2 - 2x \cos 5\pi/6 + 1.$$

$$\therefore x^6 + 1 = (x^2 - 2x \cos \pi/6 + 1) (x^2 - 2x \cos 3\pi/6 + 1) (x^2 - 2x \cos 5\pi/6 + 1)$$

$$\stackrel{r=0}{=} \prod_{r=0}^2 (x^2 - 2x \cos \frac{(2r+1)\pi}{6} + 1)$$

Now (ii) $x^9 + 1$.

$$x^9 + 1 = (x+1) \prod_{r=0}^3 (x^2 - 2x \cos \frac{(2r+1)\pi}{9} + 1)$$

(iii) $x^{13} + 1$,

$$x^{13} + 1 = (x+1) \prod_{r=0}^5 (x^2 - 2x \cos \frac{(2r+1)\pi}{13} + 1)$$

$$(iv) x^{20} + 1 = \prod_{r=0}^9 (x^2 - 2x \cos \frac{(2r+1)\pi}{20} + 1)$$

Resolve into factors; the Expression:

Q. Factorise the exprs

→ consider the eqn $x^n - 2x^n \cos n\theta + 1 = 0$.

$$x^n - 2x^n \cos n\theta + 1 = 0$$

$$x^{2n} - 2x^n \cos n\theta + \cos^2 n\theta = -1 + \cos^2 n\theta$$

$$x^{2n} - 2x^n \cos n\theta + \cos^2 n\theta = -\sin^2 n\theta$$

$$\Rightarrow (x^n - \cos n\theta)^2 = (\sin n\theta)^2$$

$$\Rightarrow x^n - \cos n\theta = \pm i \sin n\theta$$

$$= \cos(n\theta + 2r\pi) \pm i \sin(n\theta + 2r\pi), r \in \mathbb{Z}$$

$$\Rightarrow x = \frac{\cos(n\theta + 2r\pi)}{n} \pm i \frac{\sin(n\theta + 2r\pi)}{n} \quad \text{--- (1)}$$

In (1), Put $r = 0, 1, 2, \dots, n-1$.

$$r=0, \quad x = \cos 0 \pm i \sin 0$$

$$r=1, \quad x = \cos\left(0 + \frac{2\pi}{n}\right) \pm i \sin\left(0 + \frac{2\pi}{n}\right)$$

$\vdots$

$$r=n-1, \quad x = \cos\left(0 + \frac{(2n-1)\pi}{n}\right) \pm i \sin\left(0 + \frac{(2n-1)\pi}{n}\right)$$

There are $2n$ solutions of the given eqn.

Corre. $r=0$, the solutions are $x = \cos 0 + i \sin 0$ and $x = \cos 0 - i \sin 0$.

The corresp. quadratic eqns are,

$$(x - \cos \theta)^2 - i^2 \sin^2 \theta = (x - \cos \theta)^2 + \sin^2 \theta$$

$$\text{for } n=1, 2, 3, \dots, n-1, \quad = x^2 - 2x \cos \theta + 1$$

we've, $x^2 - 2x \cos(\theta + \frac{4\pi}{n}) + 1, \dots, x^2 - 2x \cos(\theta + \frac{2(n-1)\pi}{n}) + 1$
as factors. respectively.

Also on multiplying together these n factors we see that the coeff of x^{2n} in their product is unity. which is also the coeff of x^{2n} in the given expression.

So no other numerical factor need be included in the product.

$$x^{2n} - 2x^n \cos n\theta + 1$$

$$= (x^2 - 2x \cos \theta + 1) (x^2 - 2x \cos(\theta + \frac{2\pi}{n}) + 1) \dots$$

$$(x^2 - 2x \cos(\theta + \frac{2(n-1)\pi}{n}) + 1)$$

$$= \prod_{r=0}^{n-1} (x^2 - 2x \cos(\theta + \frac{2r\pi}{n}) + 1)$$

Q. Factorise the expression $x^{2n} - 2a^n x^n \cos n\theta + a^{2n}$

consider $x^{2n} - 2a^n x^n \cos n\theta + a^{2n} = 0$.

$$x^{2n} - 2a^n x^n \cos n\theta = -a^{2n}$$

$$x^n - 2a^n x^n \cos n\theta + a^{2n} \cos^2 n\theta = -a^{2n} + a^{2n} \cos^2 n\theta$$

$$(x^n - a^n \cos n\theta)^2 = -a^{2n} (-1 + \cos^2 n\theta)$$

$$(x^n - a^n \cos n\theta)^2 = -a^{2n} \sin^2 n\theta$$

$$\Rightarrow x^n - a^n \cos n\theta = \pm i a^n \sin n\theta$$

$$x^n = a^n \cos n\theta \pm i a^n \sin n\theta$$

$$= a^n (\cos n\theta \pm i \sin n\theta)$$

$$= a^n (\cos (n\theta + 2r\pi) \pm i \sin (n\theta + 2r\pi))$$

$r \in \mathbb{Z}$

$$\Rightarrow x = a \cos \left(\frac{n\theta + 2r\pi}{n} \right) \pm i \sin \left(\frac{n\theta + 2r\pi}{n} \right) \quad \text{--- (1)}$$

In (1), Put $r = 0, 1, 2, \dots, n-1$,

$$r=0, \quad x = a (\cos \theta \pm i \sin \theta)$$

$$r=1, \quad x = a \left(\cos \left(\theta + \frac{2\pi}{n} \right) \pm i \sin \left(\theta + \frac{2\pi}{n} \right) \right)$$

$\vdots$

$$r=n-1, \quad x = a \left[\cos \left(\theta + \frac{2(n-1)\pi}{n} \right) \pm i \sin \left(\theta + \frac{2(n-1)\pi}{n} \right) \right]$$

These are the n solutions of the given eqn.

When $r=0$, solutions are $x = a \cos \theta + i \sin \theta$, $x = a \cos \theta - i \sin \theta$

corresponding quadratic factors are,

$$(x - a \cos \theta - i \sin \theta) (x - a \cos \theta + i \sin \theta)$$

$$x^2 - 2ax \cos \theta + a^2$$

Thus, for $n=1, 2, \dots, n-1$ we've the quadratic factors,

$$x^2 - 2ax \cos \left(\theta + \frac{2\pi}{n} \right) + a^2, \dots, x^2 - 2ax \cos \left(\theta + \frac{2(n-1)\pi}{n} \right) + a^2$$

respectively.

Also on multiplying together these n factors we see that coefficient of x^n in their product is unity, which is also the coeff of x^n in the original expression.

$\therefore$ no other numerical factor required.

$$\begin{aligned} \Rightarrow x^n - 2a^n x^n \cos n\theta + a^n &= (x^2 - 2ax \cos \theta + a^2) (x^2 - 2ax \cos \left(\theta + \frac{2\pi}{n} \right) + a^2) \dots \\ &\quad \left(x^2 - 2ax \cos \left(\theta + \frac{2(n-1)\pi}{n} \right) + a^2 \right) \\ &= \prod_{r=0}^{n-1} \left(x^2 - 2ax \cos \left(\theta + \frac{2r\pi}{n} \right) + a^2 \right) \end{aligned}$$

Q. Factorise $x^8 - 2x^4 \cos 60^\circ + 1$

$$x^8 - 2x^4 \cos 60^\circ + 1 = 0$$

$$x^8 - 2x^4 \cos \pi/3 = -1$$

$$x^8 - 2x^4 \cos \pi/4 + \cos^2 \pi/3 = -1 + \cos^2 \pi/3$$

$$(x^4)^2 - 2x^4 \cos \pi/4 + \cos^2 \pi/3 = -\sin^2 \pi/3$$

$$(z^4 - \cos \pi/3)^2 = -(\sin \pi/3)^2$$

$$z^4 - \cos \pi/3 = \pm i \sin \pi/3$$

$$z^4 = \cos \pi/3 \pm i \sin \pi/3$$

$$= \cos (\pi/3 + 2r\pi) \pm i \sin (\pi/3 + 2r\pi)$$

$$z = \cos \left(\frac{\pi + 6r\pi}{12} \right) \pm i \sin \left(\frac{\pi + 6r\pi}{12} \right), r \in \mathbb{Z}$$

$$z = \cos \frac{(1+6r)\pi}{12} \pm i \sin \frac{(6r+1)\pi}{12} \quad \text{--- ①}$$

In ①, Put $r = 0, 1, 2, 3$.

$$r=0, \quad z = \cos \pi/12 \pm i \sin \pi/12$$

$$r=1, \quad z = \cos 7\pi/12 \pm i \sin 7\pi/12$$

$$r=2, \quad z = \cos 13\pi/12 \pm i \sin 13\pi/12$$

$$r=3, \quad z = \cos 19\pi/12 \pm i \sin 19\pi/12$$

corresp. $r=0$, the solutions are,

$$z = \cos \pi/12 + i \sin \pi/12, \quad z = \cos \pi/12 - i \sin \pi/12.$$

corresp. quadratic factor is,

$$(z - \cos \pi/12)^2 - (i \sin \pi/12)^2 = z^2 - 2z \cos \pi/12 + 1.$$

likewise corresp. to $r=1, 2$ and 3 , the quad. factors are,

$$z^2 - 2z \cos 7\pi/12 + 1, \quad z^2 - 2z \cos 13\pi/12 + 1, \quad z^2 - 2z \cos 19\pi/12 + 1.$$

resp.

$$\therefore x^3 - 2x^4 \cos \pi/3 + 1 = \frac{3}{11} (x^2 - 2x \cos(\frac{6\pi+1}{12}) \pi + 1) \quad x \geq 0$$

HW
Q. $x^{10} - 2x^5 \cos \pi/3 + 1$.

Q. Factorise $x^{2n} - 1$, where n is a positive integer.

$$x^{2n} - 1 = 0$$

$$x^{2n} = 1 = \cos 0 \pm i \sin 0$$

$$x^{2n} = \cos 2r\pi \pm i \sin 2r\pi, \quad r \in \mathbb{Z}$$

$$x = \cos \frac{r\pi}{n} \pm i \sin \frac{r\pi}{n}$$

Put $r = 0, 1, 2, \dots, n$

$$r=0, \quad x = \cos 0 \pm i \sin 0 = 1 = 1$$

$$r=1, \quad x = \cos \pi/n \pm i \sin \pi/n$$

$$r=2, \quad x = \cos \frac{2\pi}{n} \pm i \sin \frac{2\pi}{n} \quad 0 \leq r < n$$

$$r=n, \quad x = \cos \pi \pm i \sin \pi = -1$$

Corresp, $r=0, n$ we've the linear factors $x-1, x+1$.

$\therefore$ Quadratic factor is, $(x^2 - 1)$.

When $r \neq 1$, the linear factors are,

$$1 - \cos \frac{\pi}{n} - i \sin \frac{\pi}{n} \quad \text{and} \quad 1 - \cos \frac{\pi}{n} + i \sin \frac{\pi}{n}$$

$\therefore$ quadratic factor is,

$$x^2 - 2x \cos \frac{\pi}{n} + 1.$$

Now for $r = 2, 3, \dots, n-1$, we've the quadratic factors,

$$x^2 - 2x \cos \frac{2\pi}{n} + 1, \quad x^2 - 2x \cos \frac{3\pi}{n} + 1, \quad \dots, \quad x^2 - 2x \cos \frac{(n-1)\pi}{n} + 1.$$

$$\therefore x^{2n} - 1 = (x^2 - 1) (x^2 - 2x \cos \frac{\pi}{n} + 1) \dots (x^2 - 2x \cos \frac{(n-1)\pi}{n} + 1)$$

$$= (x^2 - 1) \prod_{r=1}^{n-1} (x^2 - 2x \cos \frac{r\pi}{n} + 1)$$

$$\underline{\underline{x^{2n} - 1}}$$

* 1.6. Resolve into real factors? i) $x^4 + x^3 + x^2 + x + 1$.

$$x^4 + x^3 + x^2 + x + 1 (x-1) = x^5 + x^4 + x^3 + x^2 + x - x^4 - x^3 - x^2 - 1$$

$$= x^5 - 1$$

$$\therefore x^4 + x^3 + x^2 + x + 1 = \frac{x^5 - 1}{x - 1} \quad (i)$$

$$\text{consider } x^4 + x^3 + x^2 + x + 1 = 0$$

$$\Rightarrow x^5 - 1 = 0.$$

$$\left(1 + \frac{x^5 - 1}{x - 1}\right) = \cos 2\pi \frac{1}{5} + i \sin 2\pi \frac{1}{5}$$

$$\therefore x = \cos \left(\frac{2\pi}{5}\right) \pm i \sin \left(\frac{2\pi}{5}\right)$$

Put $n=0,1,2$.

$$n=0, \quad n = \cos \pi \pm i \sin \pi = 1$$

$$n=1, \quad n = \cos \frac{2\pi}{5} \pm i \sin \frac{2\pi}{5}$$

$$n=2, \quad n = \cos \frac{4\pi}{5} \pm i \sin \frac{4\pi}{5}$$

corresp $n=0$, we've the factor $n-1$.

For $n=1,2$, the quad. factor are,

$$n^2 - 2n \cos \frac{2\pi}{5} + 1, \quad n^2 - 2n \cos \frac{4\pi}{5} + 1.$$

$$\therefore n^5 - 1 = (n-1) \left(n^2 - 2n \cos \frac{2\pi}{5} + 1 \right) \left(n^2 - 2n \cos \frac{4\pi}{5} + 1 \right)$$

$$\frac{n^5 - 1}{n-1} = \left(n^2 - 2n \cos \frac{2\pi}{5} + 1 \right) \left(n^2 - 2n \cos \frac{4\pi}{5} + 1 \right)$$

$$\Rightarrow n^4 + n^3 + n^2 + n + 1 = \left(n^2 - 2n \cos \frac{2\pi}{5} + 1 \right) \left(n^2 - 2n \cos \frac{4\pi}{5} + 1 \right)$$

thw

Q. Factorise i) $n^6 + n^5 + n^4 + n^3 + n^2 + n + 1$

$$\text{ii) } n^{10} - 2n^5 \cos \pi/3 + 1.$$

Ans:- i) $\prod_{r=1}^3 \left(n^2 - 2n \cos \frac{2r\pi}{7} + 1 \right).$

ii) $\prod_{r=0}^4 \left(n^2 - 2n \cos \left(\frac{6r+1}{15} \right) \pi + 1 \right)$

Q. S.T $n^5 - 1 = (n-1) \left(n^2 - 2n \cos \frac{2\pi}{5} + 1 \right) \left(n^2 - 2n \cos \frac{4\pi}{5} + 1 \right).$

Ans as 1. Q*